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Holistic modelling of different agroforestry systems within the ReForest project to promote adoption in Europe

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Agroforestry has the potential to produce diverse food and income sources, provide balanced development of the region and aid in sustainable management of agriculture, natural resources and climate action. IPCC recommends agroforestry as the most promising inclusive solution on land for climate change mitigation and adaptation. It is primarily supported and recognized under the Common Agricultural Policy (CAP) in Europe, and farmers adopting it are eligible for incentives under Pillar I & II.

However, as the implementation of CAP is at the discretion of Member States, different regulations at federal state levels, adoption pathways, and lack of awareness, information, and experiences on specific agroforestry methods hinder the extensive adoption of agroforestry systems. Practitioners and legislators need examples and cases to refer to, but these can take a long time. A holistic view of the system with tangible and intangible factors must be considered at the farm and large-scale levels for practitioners and policymakers to favor adopting agroforestry systems.

We generate a comprehensive graphical conceptual model of agroforestry systems developed from a detailed literature review and consultation with experts (agroforestry practitioners, researchers, policy-makers and consultants) on agroforestry systems as shown in Figure 1. The model offers a holistic overview of agroforestry systems and their interrelated dynamic components, including tangible and intangible factors. Our model takes into consideration the farming costs, benefits and risks and how these lead to an expected net present value for farmers. Planning, establishing, and maintaining agroforestry systems is often complex and is associated with a high initial investment. Market fluctuations, price variation, changing legislation, and potential crop failures due to pests, diseases, and weather events further complicate the adoption of agroforestry systems. Risk aversion increases with high uncertainty of local landuse policies, regulations and investment for farmers contemplating adopting agroforestry systems. Our model offers a synthetic representation of the current state of knowledge of the agroforestry system, encompassing all system complexities.

Figure 1 Graphical conceptual model of agroforestry systems

We translate our conceptual understanding of the agroforestry system into mathematical formulations and program simulations of the potential futures for multiple Living Labs agroforestry systems under the ReForest project. We quantify our probability distributions from all information sources, including expert knowledge elicitation methods through stakeholder workshops. We apply a Monte Carlo simulation to generate ranges of plausible outcomes and to provide assessments of economic (farm-level), ecologic and social (large-scale-level) benefits to reveal the parameters that most impact the outcomes of the systems and require further investigation. Gathering further information on these variables can help other farmers and policymakers realize the long-term benefits of agroforestry and thus promote agroforestry adoption. These models provide farmers, policymakers and stakeholders with better-informed decision-making to support the implementation and proliferation of agroforestry across Europe.

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Keywords

business models, Policy support, Risk assessment, decision analysis, Socioeconomic status, temperate agroforestry

Additional Attachment II.

